Introduction

Life exists in every corner of the planet—from high mountains to deep oceans, and from blazing deserts to the freezing polar regions. Each animal's body, life cycle, and behavior is adapted to its particular habitat, because this maximizes its chances of survival. Plant species, too, have their own adaptations that help them thrive.

Bald eagle
A North American bird of prey, the bald eagle snatches fish from lakes.

Birds

The power of flight allows birds to reach the remotest islands, and some to live in different parts of the world in summer and winter, migrating between the two. There is almost nowhere on Earth that lacks birdlife. Here are their secrets.

- **Lightweight bones**
  Most bird bones are hollow, reinforced by bony struts.
- **Flight feathers**
  Wing and tail feathers provide lift and steer the bird in fight.
- **Warming feathers**
  Two layers of body feathers keep the bird's skin warm.
- **Efficient lungs**
  Bird lungs are far more efficient than mammals', giving them the oxygen they need for energetic flight.

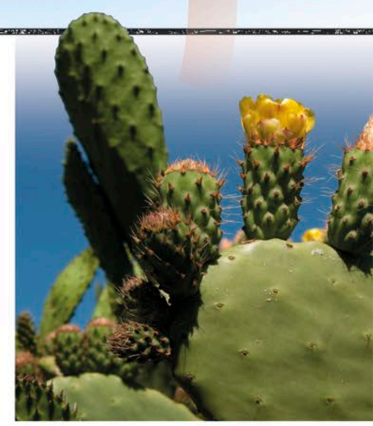
Marine animals

Living in water gives more support than living on land, so many sea creatures survive without strong skeletons. Sea water carries clouds of microscopic life-forms and dead matter, and many sea animals can afford to give up moving from place to place, fix themselves to the seabed, and "filter feed" by grabbing these passing pieces of food.

- **Gills**
  Sea mammals must surface to breathe, but fish take oxygen directly from the water using their gills.
- **Smooth shape**
  Fast-moving marine animals have a streamlined body, which helps them move through the water easily.
- **Buoyancy aid**
  Some fish have an air-filled "swim bladder" to help control buoyancy.
- **Bioluminescence**
  It is dark in the ocean depths. Many deep-sea animals produce light by chemical reactions in their bodies.

Coral
Tropical coral reefs are giant growths of filter-feeding life-forms on the seabed.

Desert cacti
The waxy, fleshy bodies of these desert plants store water. The leaves are reduced to spines, which lose less water to the air. The roots of a cactus may spread out over a wide area, to absorb as much water as possible.



Spineless cactus
A spineless variety of the prickly pear.